

Friday 5/11/2022

Midterm

Duration: 90 minutes

Name:

Student No:

P1 [20 points] Construct a DFA for the languages

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| (a) $\{w \mid w \in \{0,1\}^* \text{ starts and ends with the same symbol and has at most two 1s. }\}$ | (b) $\{w \mid w \in \{a,b\}^* \text{ where the number of } as \text{ is not a multiple of 3}\}$ |
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P2 [20 points] Construct an NFA for the languages below. Use as few states as you can. Fewest gets full point, every extra state halves your point.

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| (a) $\{w \mid w \in \{a,b,c\}^*, w \geq 3, \text{ and the last three symbols of } w \text{ are the same.}\}$ | (b) $\{w \mid w \in \{a,c,g,t\}^* \text{ starts and ends with } a, \text{ contains exactly two } as, \text{ and no consecutive symbols are the same }\}$. |
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P3 [20 points] Prove that the language $\{a^m b^n \mid \text{where } n \text{ divides } m.\}$ is not regular.

P4 [20 points] Give the CFG and PDA for the following language: $\{a^k b^m c^n \mid k, m, n \geq 0, k + n = m\}$

CFG(10pts):

PDA(10pts):

P5 [20 points] Prove that the following language is not context-free: $L = \{0^m 1^n 0^m 1^n \mid m, n \geq 0\}$